1 · Job position
Site Reliability Engineer (SRE): software engineers who run services and build systems that replace manual operations work

2 · Organization and workload
Google; SRE started there in 2003
Services SRE runs: Search, Ads, Gmail, Photos
Hundreds of on-call rotations; SRE is separate from product teams

3 · Operating system
Linux kernel underpins much of Google production
Internal Linux distribution; gLinux Rodete (Debian-based) on Googler machines
Borg: cluster OS that runs huge numbers of Linux containers

4 · Hardware and infrastructure
Google-designed datacenters: custom power, cooling, network and compute
Jupiter Clos network inside datacenters; B4 backbone between them
Thousands of machines and disks fail per cluster each year

5 · Software, tools and applications
Blaze/Bazel hermetic builds for C++, Java, Python, Go, JavaScript
Google Front End (GFE) terminates and routes traffic to most services
Practice lab (assumption): Ubuntu VM, Git, Bash/Python, Docker, Prometheus, minikube

6 · Required knowledge and skills
Reliability practice: SLOs, toil limit of 50%, blameless postmortems
OS concepts: scheduling, virtual memory, namespaces and cgroups
Hands-on preparation (assumption): Linux shell, containers, metrics, incident write-up